



MEET THE HOUNDS

CERBERUS

IoT Discovery Squad

Cerberus Hounds · Category: Discovery

“Find the unmanaged devices at the edge.”

The many-headed guardian — Cerberus is a four-agent pack watching the gate where unmanaged devices slip in.

THE PROBLEM

Unmanaged IoT and OT devices are invisible to agent-based tooling — no agent, no entry in the CMDB, no owner. They sit at the network edge as soft targets: cameras, controllers, printers, lab gear. You can't protect what nothing in your stack admits exists.

WHAT IT DOES

Cerberus is a four-stage agent squad — **Discovery** → **Expansion** → **Cross-Reference** → **QA** — that fuses web-UI and banner text, UPnP/SSDP, mDNS, OT/ICS protocol, and OUI vendor signals into a **confidence-scored** inventory of unmanaged devices. Each find is tagged **IoT:<name>**, and the squad learns new detections as it goes.

KEY CAPABILITIES

- **Confidence-scored fusion engine** — web-UI and banner text, UPnP/SSDP, mDNS, OT/ICS protocol, and OUI vendor signals combine into one High/Medium/Low verdict per device. Five weak signals become one strong answer.
- **OT/ICS protocol class** — industrial protocols are recognized as their own device class, so a PLC is classified as OT gear instead of being lumped in as “unknown device.”
- **OUI vendor and prefix mapping** — MAC-derived vendor evidence maps to device class, adding a signal that works even when a device refuses to answer anything else.
- **Managed-device down-weighting** — hosts your tooling already manages are down-weighted, keeping the list focused on the truly unmanaged edge instead of re-discovering your own servers.
- **Per-device signal breakdown** — every verdict shows exactly which signals drove it, so network teams can verify before they isolate rather than acting on faith.
- **Gated “IoT:<name>” tags** — discovered devices become human-approved, routable inventory the rest of the pack can prioritize and protect.

HOW IT WORKS

Cerberus runs against the local navi.db built from your Tenable data — the **vulns** and **assets** tables plus the cert-derived device cache from Certania. All writes are proposed, human-approved, and logged.

WHY IT'S DIFFERENT

- **Confidence scoring instead of brittle single-signal detection** — one banner match isn't a verdict; five agreeing signals are.
- **Signal transparency** — every device shows its evidence breakdown, so network teams can verify before they isolate.
- **OT-aware** — industrial protocol classes are first-class citizens, not lumped in with smart lightbulbs.
- **A squad that learns** — the QA stage feeds new detections back into discovery.

PROOF POINTS

- Runs its four-stage sweep across a reference environment of 268 mixed assets — including IoT/OT gear — scoring each candidate device High, Medium, or Low confidence with the signals shown.

Honest caveat: detection quality depends on what your scans can observe at the edge — blind segments and uncredentialed hosts are called out, not hidden. Confidence scores are evidence summaries, not certainties.

WORKS BETTER WITH

Cerberus feeds **Anubis** — so lab and OT devices can be down-rated (or up-rated) to their true criticality — and **Fenrir**, where weak-auth edge devices become attack-path entry points. It consumes **Certania's** cert-derived device cache.

WHO IT'S FOR

OT and IoT security teams who need an inventory they can defend; network teams hunting the unmanaged edge; vuln management extending coverage beyond agents.

CALL TO ACTION

Turn Cerberus loose on your edge — three heads see what one scanner misses.